Mastering the Dynamics of Innovation

How Companies Can Seize Opportunities in the Face of Technological Change

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CHAPTER 2

Dominant Designs and the Survival of Firms

CHAPTER 1 DESCRIBED HOW THE TYPEWRITER INDUSTRY attracted a great number of entrants between 1885 and 1915. With the potential of the new invention made clear through expanding sales, entrepreneurs—many of them renegades from the Remington and Caligraph organizations-appeared with a number of uniquely designed machines. This is fairly typical in new industries of assembled products: a pioneering firm gets the ball rolling with its initial product, a growing market begins to take shape around that product, and new competitors are inspired to enter and either expand the market further or take a chunk of it with their own product versions. At this embryonic stage, no firm has a "lock" on the market. No one's product is really perfected. No single firm has mastered the processes of manufacturing, or achieved unassailable control of the distribution channels. Customers have not yet developed their own sense of the ideal product design or what they want in terms of features or functions. The market and the industry are in a fluid stage of development. Everyone-producers and customers-is learning as they move along.

This environment is conducive to market entry by many firms as long as capital and technical barriers are not too high, as was the case with typewriters in the late nineteenth century. One did not need to be a rocket scientist to develop a new typewriter design. A good hand with mechanical gadgets was what was

required, and partners like Remington were around to handle the technical- and capital-intensive business of manufacturing that new design.

At this stage in a product's evolution, both producers and customers are experimenting. Even as new companies enter with uniquely designed products, established firms are busy perfecting their original designs and introducing new models; and customers are not yet so wedded to any particular design or company that they will not experiment with something new. Industry standards at this stage are usually rudimentary, when they exist at all.

Within this rich mixture of experimentation and competition some center of gravity eventually forms in the shape of a dominant ***product design.*** Once the dominant design emerges, the basis of competition changes radically, and firms are put to tests that very few will pass. Before long, the ecology of competing firms changes from one characterized by many firms and many unique designs, to one of few firms with similar product designs.

This chapter examines that changing ecology of firms and considers 1) the idea of dominant design, 2) the conditions under which it occurs, 3) how the emergence of a dominant design affects the pace of innovation and the structure of an industry, and 4) the extent to which a dominant design can be recognized when it first appears. We consider these issues in terms of several important modern industries.

WHAT IS A DOMINANT DESIGN?

A dominant design in a product class is, by definition, the one that wins the allegiance of the marketplace, the one that competitors and innovators must adhere to if they hope to command significant market following. The dominant design usually takes the form of a new product (or set of features) synthesized from individual technological innovations introduced independently in prior product variants. Looking back over the early history of the typewriter industry, we can see that the Underwood Model 5 did this and emerged as the dominant design. It was enormously well received by the marketplace, and the subsequent introduction of copy-cat models by its major rivals (the Remington Monarch in 1900 and the L.C. & Smith Brothers Model 8 in 1907), both of which trailed Underwood's sales by a wide margin, confirmed its

dominance. With its single QWERTY keyboard, visible type, tab feature, shift-key capitalization, carriage cylinder, and so forth, this particular design brought together a number of market-proven innovations into a single machine and very quickly came to command the typewriter industry. It remained the dominant design for decades, defining how the typewriter was ***supposed to look and operate*** in the minds of both typists and other typewriter producers. Any firm that wanted to offer a keyboard with an innovative arrangement of letters, or that wanted a circular type wheel (like the old Burt design), did so at its peril; it might capture some small niche of the market where those features had demonstrated merit, but it could abandon any hopes of being a mainstream producer with those sorts of designs.

The IBM PC format quickly became the dominant design in its market, and in retrospect we can see how its success followed a pattern similar to that of the Model 5—doing in the digital age what Underwood successfully managed in an age of mechanization. The PC, like the Model 5, offered the market little in the way of breakthrough technology, but it brought together familiar elements that had proven their value to users: a TV monitor, standard disk drive, QWERTY keyboard, the Intel 8088 chip, open architecture, and MS DOS operating system. Together, these elements came to define the ***idea*** of the personal computer for at least 80 percent of the market. By the early 1990s, software that added a user-friendly interface (Windows™) into this design attempted to bridge that last important chasm between the Apple Macintosh and PC poles of the user market.

A dominant design embodies the requirements of many classes of users of a particular product, even though it may not meet the needs of a particular class to quite the same extent as would a customized design.' Nor is a dominant design necessarily the one that embodies the most extreme technical performance. It is a so-called satisficer of many in terms of the interplay of technical possibilities and market choices, instead of an optimizer for a few.

A dominant design drastically reduces the number of performance requirements to be met by a product by making many of those requirements ***implicit*** in the design itself. Thus, few today would ask if a car had an electric starter and electric windshield wipers, or whether a typewriter could produce upper- and lowercase let-

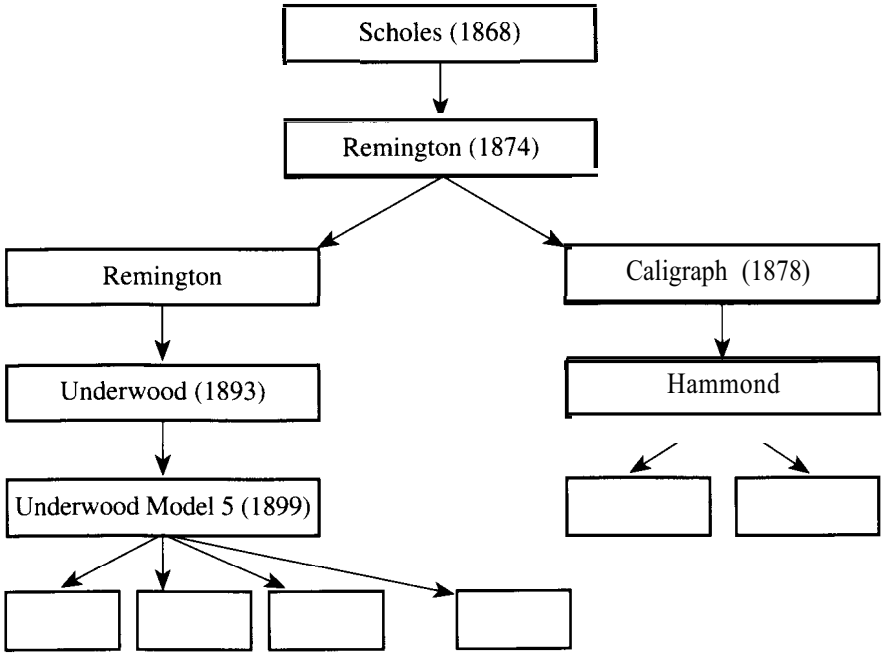
ters, or whether a personal computer had a built-in disk drive, though these were unique features in models that preceded the dominant design. Today, these features are implicit in designs that the market expects and that all producers find themselves compelled to emulate. They are no longer serious issues nor are they advertised as advantages of one or another manufacturer's product. They are subsumed within the popularly accepted design.

HOW DOES A DOMINANT DESIGN OCCUR?

If the emergence of a dominant design is the landmark event for an industry as hypothesized, then managers need to understand how dominant designs occur. It would be tempting to think that there is some predetermination to the emergence of a dominant design-that automobiles with internal combustion engines were somehow exactly what the gods of transportation always meant for us to have, and that earlier experiments with electric- and steam-powered cars were misguided aberrations destined to go nowhere. The emergence of a dominant design is not necessarily predetermined, but is the result of the interplay between technical and market choices at any particular time. This is easily illustrated by the standard keyboard that found its way into the first Remingtons, the Underwoods, and virtually all successor typewriters, computers, and other keyboards to this day. From a practical standpoint, dozens if not hundreds of different keyboard designs might have been as effective ergonomically. What Sholes, Remington, and hundreds of others settled on was the product of the experiments, technical possibilities, individual choices, proprietary positions, and-to some extent-sheer inertia. Equally, the persistence of dominant designs illustrates the momentum of both established practice and complementary assets such as-in the case of typewriters-typing skills and training.

Indeed, this notion of dominant design as successful evolution can be represented using Kim Clark's design hierarchies approach; a simplified hierarchy drawn from the history of the typewriter is illustrated in Figure 2-l. A "technological trajectory" (two are shown here) is the path of progress established by the choice of a core technical concept at the outset. Decisions about the product, constrained by prior technical choices and by the evolution of customer choices, influence these various trajectories.

FIGURE 2-l. **Design Hierarchies and Dominant Design**



The idea of a dominant design is conceptually broader than technical competition and progress. Factors other than technology come into play; chief among these are collateral assets, industry regulation and government intervention, strategic maneuvering by individual firms, and communication between producers and users.

Collateral Assets

Collateral assets-or ***co-specialized*** assets, as described by Teece[2]—seem to have a two-way relationship with the emergence of a dominant design. A firm in possession of collateral assets such as market channels, brand image, and customer switching costs will have some advantage over its competitors in terms of enforcing its product as the dominant design. The experience of IBM in personal computers is a case in point. There were plenty of firms with PCs on the market in 1981, the year in which IBM first offered its personal computer. To the buying public, the name IBM had

tremendous brand value; as a huge firm, its entry meant that replacement parts and service would be available and that applications software would begin to appear, encouraged by industry standards that would conform to IBM's machine.

Industry Regulation and Government Intervention

Industry regulation often has the power to *impose* a standard and thus define a dominant design. For instance, the FCC's approval of the RCA television broadcast standard worked to the advantage of RCA by establishing its design as the standard for the emerging television industry. Today, efforts by governments around the world to define standards for high-definition television-standards that will either favor or undermine the interests of their domestic producers-will certainly have an impact on dominant design in that field.

Strategic Maneuvering at the Firm Level

The product strategy followed by a firm relative to its competitors may determine which firm's product design becomes dominant. This is what Cusumano, Mylonadis, and Rosenbloom have called "strategic maneuvering."[3] In their study of videocassette recorder standards, these researchers determined that one reason the VHS system backed by JVC won market dominance over Sony's Betamax was the different strategies followed by these two firms. While JVC followed a "humble" strategy of establishing alliances first in Japan and then in Europe and the United States, Sony chose to go it alone-stressing its reputation and deliberately avoiding alliances or contracts that would impinge on its vertical integration for this important new market. According to Cusumano et al., it was primarily JVC's strategy, and neither the technological advantages of VHS, the firm's collateral assets, nor government regulation that made VHS the dominant design. In fact, VHS was technically inferior in many ways to its Betamax rival. Apple Computer followed a similar tactic with its Apple II, encouraging many suppliers of compatible peripheral equipment to invest in its growth. With the more exclusive architecture of the Macintosh, Apple abandoned this idea to its detriment.

Communication between Producers and Users

Finally, the way each firm manages communication with its customers many have a significant effect on its ability to impose a dominant design. This is a case of market learning. Staying close to customers makes it possible for producing firms to observe how their evolving products are actually being used, how they are succeeding or failing to satisfy customer requirements, and how design changes might close the gap between product capabilities and user requirements. Close contact with users during the period of experimentation helps firms determine which product features are truly important to customers. Managing this producer-user interface may take the form of close ties to "lead users,"[4] users' associations, and industry groups. Conversely, Christensen has shown that customers for entrenched products can influence a firm not to change when change is called for, increasing their resistance and vulnerability to technological progress.

DOMINANT DESIGN, COMPETITION, AND INDUSTRY STRUCTURE

Economists have been comfortable with the notion that the creative synthesis of a new product innovation by one or a few firms results in a temporary monopoly situation, high unit profit margins and prices, and sales of the innovation in those few market niches where it possesses the greatest performance advantage over competing alternatives. This is in line with Schumpeter's path-breaking "creative destruction" model and subsequent studies on the economics of innovation.[5] As demand and production grow, and as more applications are found for the innovation, many new firms enter the market with variations of the product. For example, early versions of the automobile included steam and electric vehicles as well as the now familiar internal combustion engine; there were literally dozens of these auto producers in the United States alone during the early auto days. Typewriter companies, we recall, entered by the dozens with curved keyboards, double keyboards, and other exotica. There were more than 30 of these companies doing business in the United States during the formative years of the typewriter industry.

Burton Klein suggested a profound connection between industry structure and technological change in his seminal work on dynamic economics.[6] Klein portrayed each firm's investments and product introductions as experiments providing corrective and stimulating feedback about product and market requirements. Thus

the earliest period in the development of a product line or industry in which few firms participate would necessarily be a period of relatively slow technical progress and productivity advance. As larger numbers of firms entered the arena, broadening the range of experimentation and the definition of the product technology, Klein expected greater innovation with correspondingly greater technology progress and productivity advance.

Finally, as a few firms came to dominate the industry with superior product technology and productivity, both experimentation and progress would be expected to slow. The renewal or broadening of competition would be required for rapid progress to recur. In reviewing earlier work, Klein found no case in which a major advance, one that established a new and more rapid trajectory for technological progress, came from a major firm in the industry in question (an unsettling point I examine in detail later). From this evidence he concluded that the process of moving from a dynamic organization to a static one, from a period of rapid organizational learning to a period of slow or no progress, appears to be irreversible.

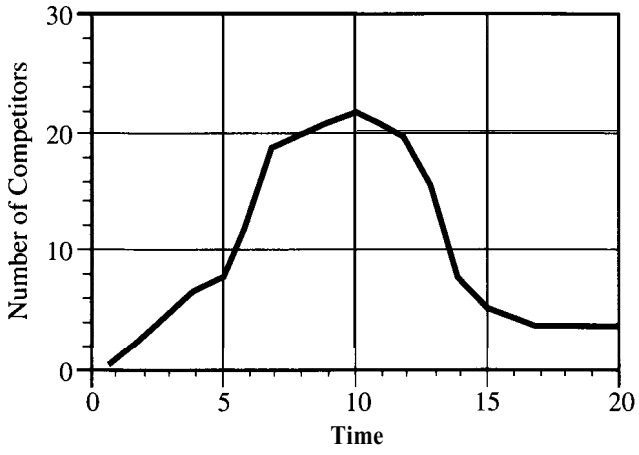
It is common to think of innovative entrants to a young industry as being small firms. This is not entirely the case. Remington was a major industrial company by the standards of the day when it went into the typewriter business. So was IBM in 1981 when it decided to enter the PC business and earlier when it entered the electric typewriter business. This is an important point we will return to in Chapter 9. The appearance of a dominant design shifts the competitive emphasis in favor of those firms-large or small—that are able to achieve greater skills in process innovation and integration and with more highly developed internal technical and engineering skills. Once the dust has settled on the contest for *product* innovation, then competitive engagement shifts to a new battleground: *process* innovation. When the marketplace decides that the QWERTY keyboard, or some other design standard, is what it wants, then innovators start figuring out how to make that peculiar keyboard as efficiently as possible; and some firms will be better able to do that than will others.

Firms that are unable to make the transition toward greater process innovation are unable to compete effectively and very often fail. Ones that possess special resources may successfully merge with the ultimately dominant firms. Overall, the inability to change organizational structure and practices in tandem with the evolution of technology in the industry is a major source of failure.

One might hypothesize that the peak of the total population curve for competition in any assembled product industry will occur within a year or two of the time when a dominant design emerges in that industry. In the course of this chapter, evidence from a number of other industries-old and new-addresses this point. Prior to the appearance of a dominant design, one would expect to see a wave of entering firms with many varied, experimental versions of the product, just as we did with the typewriter, the personal computer, and the early automobile. Following the emergence of the dominant design, the expectation, given this hypothesis, would be a wave of exits by competing firms and a general consolidation of the industry. This is represented in Figure 2-2. Thus the appearance of a dominant design in year 10 would mark an important change in the evolution of an industry, and effective competition would begin to take place on the basis of cost and scale as well as on the basis of product performance.

If this hypothesis is correct, we will have observed a repeatable pattern in the dynamics of innovation and change that is useful for both managers and scholars. The following developments might be hypothesized as gaining strength in the wake of the dominant design.

FIGURE 2-2. Dominant Design and Numbers of Competing Firms



Enforcement of Standards

A dominant design has the effect of enforcing or encouraging standardization so that production or other complementary economies can be sought and perfected. Effective competition then shifts from innovative approaches to product design and features, to competition based on cost and scale as well as on product performance.

Change in the Pace and Direction of Innovation

Once accepted, a design can have a profound impact on both the direction of further technical advance and the rate of that advance. The wide acceptance of the QWERTY keyboard--one component of the dominant typewriter design-made further innovations in keyboard design off-limits. Innovators might tinker with the peripheral keys, or add new keys like the / (slash) or the + (plus), but the rest became untouchable; if keyboard changes were to be made, they would have to be limited to other dimensions of the keyboard just as the hidden mechanisms that controlled its movements were.[7]

The pace at which major innovations find their way into the product slows down considerably after the emergence of the dominant design. Perhaps operating on the adage that you do not take a well person to the surgeon, firms turn their energies away from the innovation of product features and toward the innovations that will lead them to cost or quality advantages on what has become a fairly standardized product. The illustrations that follow make this point clear.

EVIDENCE AND EXAMPLES

We turn now to a discussion of some specific examples that exhibit the waves of change suggested by this model of dominant design and the survival of firms: the manual typewriter industry, the auto industry, the electronic calculator, integrated circuits, television and television picture tubes, Winchester disk drives, and the massively parallel supercomputer. In general, each of these industries exhibits a similar pattern of firm entries and exits.

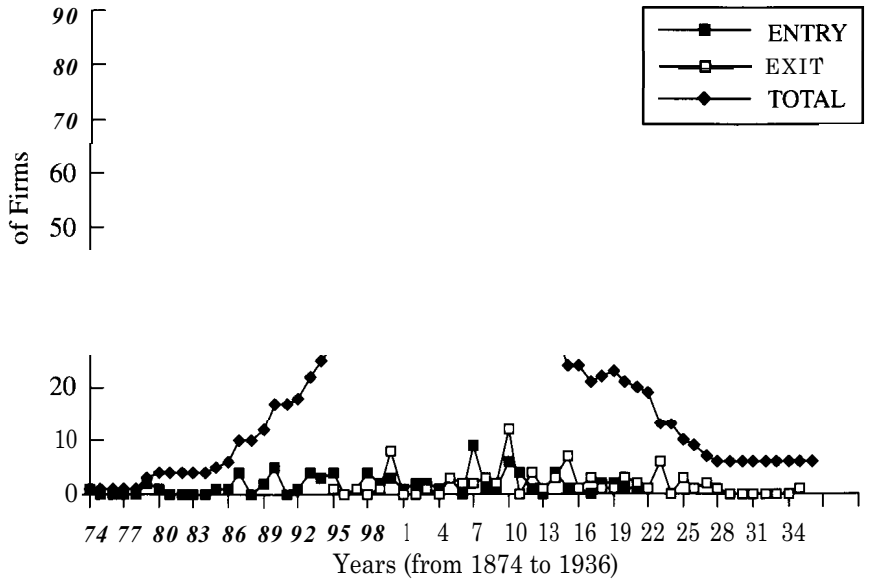
At the dawn of a new industry, firms (mostly small, entrepreneurial firms) enter at a moderate pace. Later, a rapid wave of entry occurs, raising the number of firms in the industry steeply.

After a dominant design is established, the total number of firms declines steadily until it stabilizes with a few large firms. Very often the successful firms are those that entered the industry in its early stages.

Typewriters Revisited

Growth in the typewriter industry was initially slow, probably because no one had mastered the typing skills needed to capture the full value of the new machine. By 1885, four new competitors had appeared, breaking Remington's monopoly, and the typewriter was gradually accepted by the public. By the turn of the century over 30 firms had machines in the U.S. market, but until that time typewriter manufacturers offered products with few standardized characteristics. Royal appeared in 1904, and by 1909, as seen in Figure 2-3, the number of competing firms had increased to 40. Underwood introduced its Model 5 in 1899. The visible writing of the Model 5 allowed the typist to see what he or she had actually

FIGURE **2-3. Number of Firms in the U.S. Typewriter Industry**



Source: Based on data in George Nichols Engler, ***The Typewriter Industry: The Impact of a Significant Technological Innovation*** (Los Angeles: University of California at Los Angeles, 1970).

typed as the keys struck the page. It was the first to have a tabulator-making columnar presentations much simpler-and it was able to cut stencils and make good copies. These valued features won Underwood a large share of the commercial office market. And as more people learned to use the Model 5, it formed their expectations of what a typewriter should be. From that point on, the essential features of the typewriter were set in the Underwood machine. It became the dominant design-the model of "typewriterness" that the public expected and that others would eventually follow.

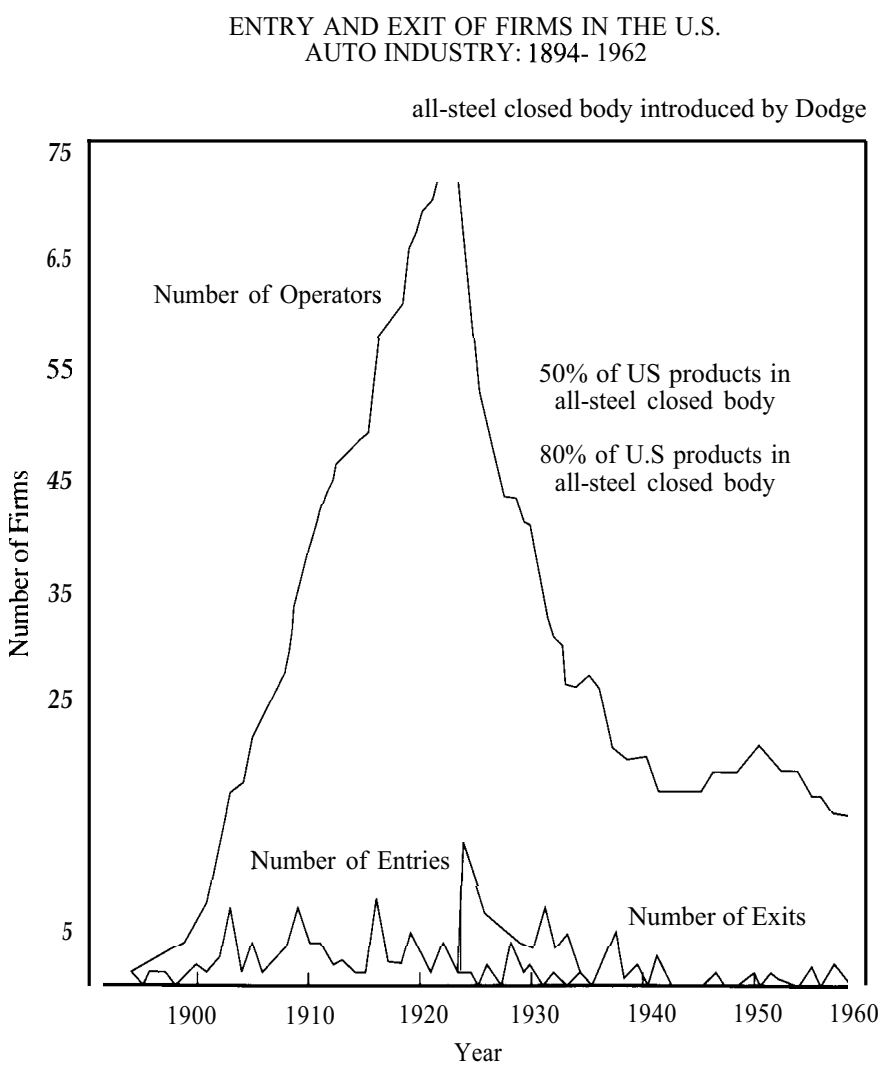
The introduction of the Underwood Model 5 marked a parting of the waters in the crowded typewriter industry. As Figure 2-3 indicates, the end of rapid growth in the number of competing firms occurred shortly after the introduction of the Model 5; after 1907, the number of firms in the industry began a slow and irreversible decline. By 1940, only five producers had any standing in the market: Remington, Royal, Smith, and Underwood each with 20 percent market share, and IBM with about 10 percent. Others, mostly foreign firms, had the remaining 10 percent.[8] More than 90 percent of the firms that had entered the industry had disappeared, either through bankruptcy or, in a few cases, merger. Only a few of the early innovative entrants survived. Except for a variety of embellishing features, little would change in the product or the competitive situation until the introduction of the electric typewriter. Even then, the electric typewriter represented less of a change for the user than did the advent of the first manual machine at the turn of the century. It remained for the development of the personal computer to provide an example of similar broadening of the market.

We can hypothesize that the emergence of a dominant design very suddenly shifted the basis of competition away from the development of unique product features to other capabilities such as low-cost manufacturing, marketing, service, or some other factors; indeed, George Engler, whose study of the industry has been cited here and in Chapter 1, reached that conclusion.

Automobiles

More than 100 firms entered and participated in the U.S. auto industry for a period of five years or longer, as presented in Figure 2-4. The wave of entry began in 1894 and continued through 1950; the wave of exits began in 1923 and peaked only a few years later, although it has continued until the present day.

FIGURE 2–4. **Number of Firms in the U.S. Automobile Industry**



Source: James M. Utterback, "Innovation and Industrial Evolution in Manufacturing Industries," in Bruce Guile and Harvey Brooks, eds., reprinted with permission from Technology and *Global Industry* (Washington, D.C.: National Academy Press, copyright 1987), based on data from R. Fabris, "Product Innovation in the Automobile Industry," Ph.D. diss., University of Michigan, 1966.

As we would hypothesize, entry of firms into the industry began slowly, but accelerated rapidly after 1900. From 1894 to 1918, 60 firms entered the industry and none left; participation reached a peak of 75 firms in 1923. In the next two years, 23 firms, nearly one-third of the industry, left or merged, and by 1930, 35 firms had exited. During the Great Depression, 20 more firms left. There was a brief flurry of entries and exits immediately following World War II, but the number of U.S. firms in the industry was basically stable between 1940 and 1980, when a number of foreign producers set up production operations in North America.

A number of major product innovations appear within this pattern of entries and exits. We do not have data on innovations for the 1894-1918 period, but we assume that during these formative years product innovations were frequent and substantial. The year 1923, when the number of competing firms reached its peak, was the year that Dodge introduced the all-steel, closed body automobile-an important innovation. This new body format dramatically improved the strength and rigidity of the chassis, and at the same time provided an opportunity for manufacturers to move away from hand forming of exterior body panels to the highly capitalized but efficient process of machine stamping.

By 1925, fully half of U.S. auto production was all-steel, closed, body cars; and by 1926, 80 percent of all automobiles were of this type.[9] Exits from the industry began and picked up speed rapidly around this time, and the number and rate of product innovations declined markedly. New concepts that did come along in product accessories and styling were tested in the low-volume, high-profit luxury automobile. Conversely, incremental innovations were more commonly introduced in lower-priced, higher-volume product lines. General Motors appears to have led in both types of innovations, particularly for major product changes. Between 1919 and 1962, the U.S. auto industry witnessed a declining number of major innovations on an annual basis, a decrease in the number of new entrants, and a steady withdrawal of firms from the business (see Table 2-1).[10]

The Dodge all-steel, closed body design was introduced during the period in which fully half the major innovations took place-1919-1929. While it clearly became the dominant design for the auto body and a major part of the total product package, its influence on the rate of firm exit and entry cannot be established causally. However, the link between major innovations and the decline of new entrants and increase in exits fortifies our intuition about the linkages here.

TABLE 2-1. **Major Innovations and Firm Entries/Exits from U.S. Auto Industry**

Period	*Innovations*	*New Entrants*	*Firm Exits*
1894-1918	NA	60	0
1919-1929	14	22	43
1930-1941	11	6	29
1946-1962	7	4	8

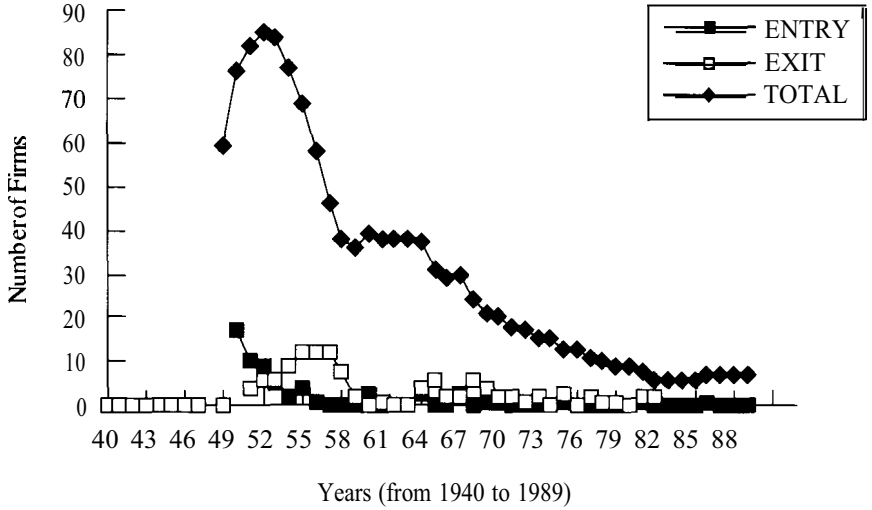
Source: R. Fabris, *Product Innovation in the Automobile Industry,* Ph.D. diss., University of Michigan, 1966, p. 92.

Televisions and Television Tubes

Research leading to the appearance of the television started several decades before the first successful results were achieved. RCA entered the industry in 1929 after David Sarnoff, impressed by a demonstration by inventor Vladimir Zworykin, decided to hire Zworykin and put him in charge of RCA's Electronic Research Group in Camden, New Jersey. Several other firms or inventor-entrepreneurs entered the infant industry during the 1930s, and all contributed to the expanding frontier of technical knowledge. Philo Farnsworth, Louis Hazeltine, American Television, and Allen DuMont are some of the most important names in the early years of television.

The industry's commercial birth can be traced back to the 1939-1940 World's Fair in New York, where millions saw television displays for the first time. For the purposes of our analysis, 1939 marks the beginning of the industry. Data in Figure 2-5 start in 1949. The figure clearly shows that the television industry conforms to the hypothesized relationships between dominant design and the peaking of competing firms in the industry. The first decade of the industry obviously witnessed a rapid increase in the number of firms making television sets. The wave of entry most likely peaked in 1950, the first year of our entry data, or one year earlier. The total number of firms steadily increased until 1952, the year in

FIGURE 2-5. Number of Firms in the U.S. Television Industry



Source: James M. Utterback and Fernando Suárez, "Innovation, Competition, and Industry Structure," *Research Policy,* vol. 22, no. 1, January 1993, p. 11.

which it peaked at 85 firms. The wave of exits started to take off in 195 1 and peaked around 1956.

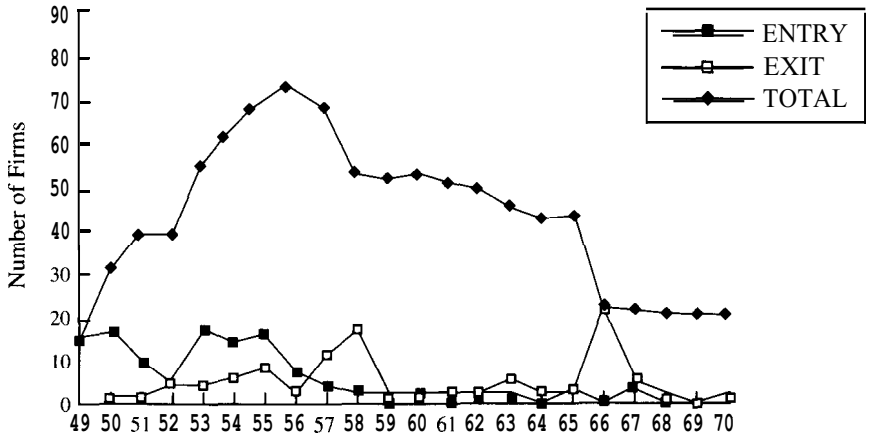
Several things happened in the early 1950s television industry that had a significant impact on the pattern of innovation and competition that was to follow. First, the uncertainty about technical standards for color broadcasting (i.e., UHF versus VHF) was resolved by the Supreme Court in 195 1. Two years later the FCC approved the National Television Standards Committee (NTSC) system, backed by a group of manufacturers headed by RCA. Several firms that had opposed the RCA technical standards dropped out of the business due to this legal ruling. Second, several features of the television set converged to form a dominant design around 1952. The most important aspect of this design was the size of the screen and therefore the characteristics of the picture tube. The first set produced by RCA had a IO-inch monochrome tube. Almost all tubes produced in the 1940s were smaller than 14 inches. RCA produced its first 21-inch and other large tubes around 1952, and these soon became the market standard. Third, during the early 1950s, RCA started licensing its television technology to other

firms. This further reduced competition and supports the idea that RCA held rights to most of the key characteristics of the product at the time.

The story of the TV picture and receiving tubes is closely related to that of the television industry itself. Figure 2-6 depicts a rapid increase in the total number of tube producers from 1949 to 1956, analogous to that previously discussed for the television industry. In 1956, four years after the peak of the television industry curve, the total number of firms in the tube industry reached its peak of 66 firms.

One of RCA's major innovations was the development of the shadow-mask color picture tube. The FCC ruled in favor of the RCA-compatible standard in December 1953, and programs were first broadcast in this format in January of 1954. Hazeltine made major contributions to brightness. Some problems in manufacture constrained the size of the tubes to 16 inches until CBS laboratories learned how to curve the mask. RCA licensed both developments. A problem with the initial tubes was that they were metallic with glass bonded to the front, and this interface proved troublesome, often separating and ruining the tube. The 21-inch, all-glass tube-first black and white, then color-eliminated this problem

FIGURE 2-6. Number of U.S. Picture Tube Producers



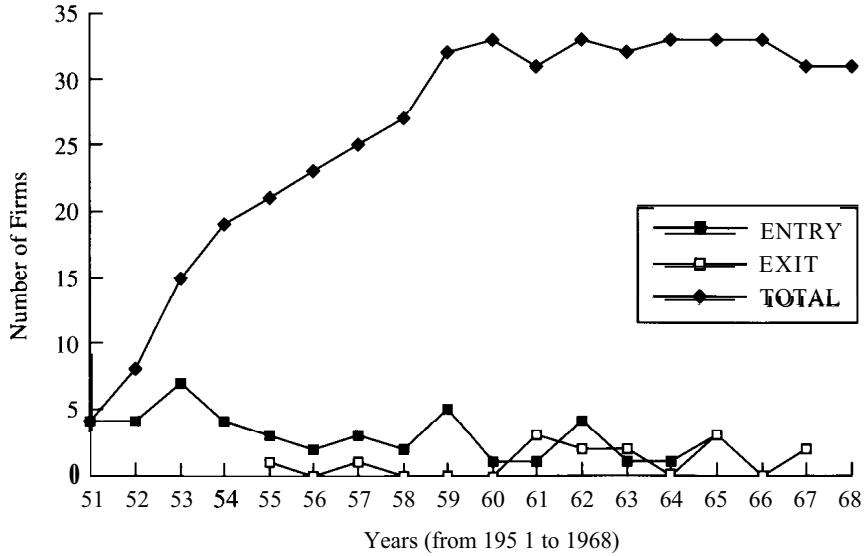
and has been by far the biggest seller since the mid-1950s. The first such tube was introduced by RCA late in 1955 and captured a significant portion of the TV tube market during the rest of the decade. The advent of reliable color tubes contributed to the difficulties that many tube producers were experiencing during the late 1950s and fostered concentration in the industry. Indeed, in May 1958, RCA publicly announced the first all-glass 21-inch color picture tube, which was to hold a large share of the color tube market."

The Transistor

According to J.E. Tilton, three firms dominated the receiving vacuum tube business in the United States in 1950: General Electric, Philco Ford, and RCA.'* Over the ensuing decade these firms spent more on R&D and received more patents than did new firms entering the industry (counting the expenditures and patents of Bell Laboratories), but a cohort of new entrants gained nearly two-thirds of the new and growing transistor market. By 1966, three of these new entrants-Texas Instruments, Motorola, and Fairchild Semiconductor-together accounted for 42 percent of the market, while the vacuum tube manufacturers' share of the transistor market declined to just slightly more than one-fourth of the total. The difference between established firms and new entrants in this business would be even more dramatic if one included IBM's production for its own use (IBM is believed to have entered production in 1961).

Figure 2-7 shows that the total number of firms in the transistor industry began to rise rapidly with its invention in 1948. This increase was virtually halted around 1959, when the planar transistor entered commercial production. Transistors produced through the planar process rapidly became the dominant design in the industry. Introduced by Fairchild Semiconductors, planar transistors presented many advantages over mesa transistor technology. In particular, the planar transistor was flat, which meant that electrical connections could be achieved by depositing an evaporated metal film on appropriate portions of the wafer. This was a great advantage over the mesa transistor, whose irregular surface dictated that electrical connections be done laboriously by hand. The planar transistor prompted a drive to produce low-cost transistors-a phenomenon typical of post-dominant design stages. The

FIGURE 2-7. **Number of Firms in the U.S. Transistor Manufacturing Industry**



Source: Based on data in E. Braun and S. MacDonald, *Revolution in Miniature: The History and Impact of Semiconductor Electronics* (Cambridge, England: Cambridge University Press, 1978); and J.E. Tilton, *International Diffusion of Technology* (Washington, D.C.: The Brookings Institution, 1971).

reader will note that exits from the industry, almost nil before 1959, became commonplace thereafter.

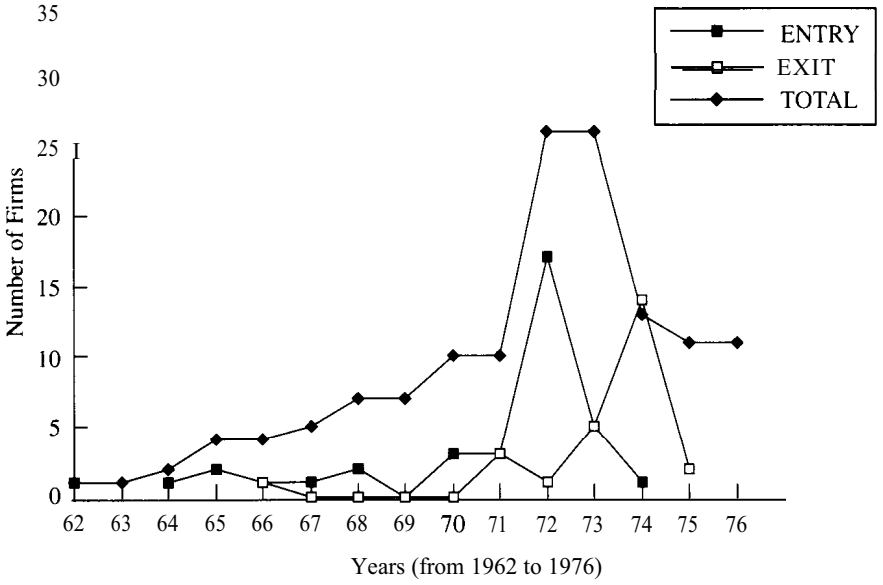
The Electronic Calculator

The American calculator industry in the early 1960s consisted of five major companies manufacturing the electromechanical machines that controlled nearly 90 percent of the market: Frieden, Monroe, Marchant, Victor, and Olivetti. Frieden, Marchant, and Monroe each had approximately 20 percent of the market, Victor a slightly smaller share, and Olivetti 10 to 15 percent. These companies were almost completely vertically integrated because of the need for a high degree of precision in the manufacture of many specialized parts. There were strong barriers to new-firm entry. By concentrating on specific segments of the market, the major companies avoided intense competition. Extensive distribution and service networks reinforced their market dominance. In addition,

the technology of electromechanical calculators had reached a high level. It was proving difficult for any firm to come up with a dramatic breakthrough that would threaten the status quo.

Initially this situation did not change when the electronic calculator entered the market in 1962. The first electronic calculators were extremely complex and expensive machines of more than 2,300 discrete parts aimed at specialized scientific and technical market segments. Figure 2-8 shows that between 1962 and 1970, 11 firms entered the industry, with 10 surviving. This wave of entry peaked in 1972 with the entry of 21 firms in a three-year period. However, exits began in 1971, following the introduction of the calculator chip, a dominant design that made unit assembly a simple matter of piecing together the chip, display device, and keyboard. The entry of semiconductor manufacturers such as Texas Instruments and Rockwell in 1972, and National Semiconductor in 1973, further precipitated the departure of firms that were largely

FIGURE 2-8. **Number of Firms in the U.S. Electronic Calculator Industry**



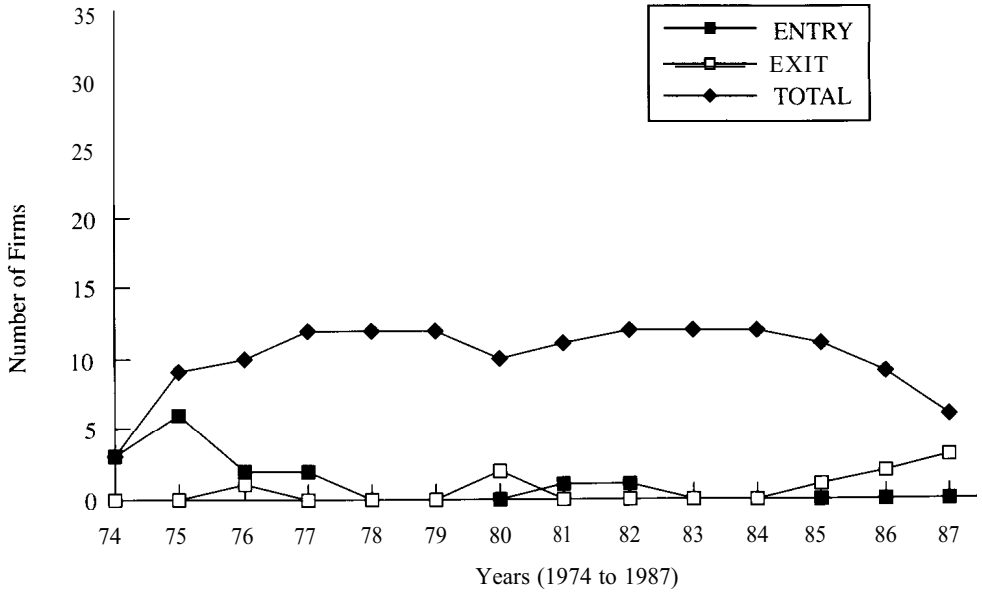
Source: **Based on data in B. Majumdar,** *Innovations, Product Developments, and Technology Transfer: An Empirical Study of Dynamic Competitive Advantage, The Case of Electronic Calculators,* **Ph.D.** diss., Case Western Reserve University, Cleveland, Ohio, 1977.

assemblers of purchased components. The industry's structure then appears to stabilize, with a few of the semiconductor makers (e.g., Rockwell) dropping out and a small number of the remaining vertically integrated companies dominating the market. Thus the appearance of a dominant design and the drive toward vertical integration that normally follows its appearance were almost concurrent in this highly compressed example.

The Integrated Circuit

The integrated circuit industry is the only one in the sample that does not clearly conform to the hypothesis regarding a collapse of the number of competitors following the emergence of a dominant product design. In fact, Figure 2-9, displaying data on U.S. firms, shows no clear peak in firms during the data period. Therese Flaherty, who has studied the integrated circuit industry, suggests that no one product of any generation can be easily considered a dominant design. The product has continued to change substantially from generation to generation, which may explain the very

FIGURE 2-9. **Number of Firms in the U.S. Integrated Circuit Industry**



Source: Dataquest and Therese Flaherty, personal communication.

broad plateau observed in total industry participation as many firms both enter and exit the industry.

There have been many generations of integrated circuits. For instance, DRAM (dynamic random access memory), the most important segment of the industry at this time, has already passed through seven generations (1K, 4K, 16K, 64K, 256K, 1MB, 4MB). Competition has been intense within and between generations, and no firm has been able to maintain a leadership position from one generation to another. Based on this industry, Fumio Kodama introduced the concept of "surf riding." If one company leads the race in, say, 4K memories, it will not lead in the 16K generation. Having lost that one, however, it may attempt to strike back in the contest for 64K memories, where the leader of the 16K generation will not likely figure as a prominent competitor. In general, American firms have been losing ground to Japanese entrants. The first three generations were dominated by American firms, but Japanese firms began taking a significant share of the market from the 64K generation onward.

Although a quick look at the integrated circuit industry suggests that the model has limited explanatory power in this case, several issues cast a shadow on such a conclusion. The first is Flaherty's observation that within each generation of integrated circuits the "enter early" strategy was a winner. Second, the production capacity of the major firms has increased relative to market demand through each generation; fewer companies are needed to satisfy demand.

Exits for firms from the American integrated circuit industry began in 1985 and increased steadily for the next two years. As new entries were nil during this period, the total number of American firms declined in 1987 to one of its lowest levels. Today the prevailing industry pattern is higher concentration and larger firms. Finally, although product innovation is still important in later integrated circuit generations, process innovation and production capabilities are increasingly critical as generations pass and greater production volumes are required of participant firms. Such production capabilities form an effective barrier to entry in the industry.

Winchester Disk Drives

The technology for Winchester disk drives was first developed by IBM in 1973 as a high-speed method of storing data. A number of competing firms emerged with similar technology and products. At

first these were designed for use with mainframe and minicomputers, but future generations of the product addressed the storage needs of the proliferating base of microcomputers. The Winchester disk drive industry attracted many startup firms, many of which enjoyed success in the early years. Between 1978 and 1983—the year in which the number of firms producing these drives peaked—sales to original equipment manufacturers grew from $27 million to $1.3 billion. Industry analysts were euphorically projecting a tripling of revenues from that level by 1987.[13]

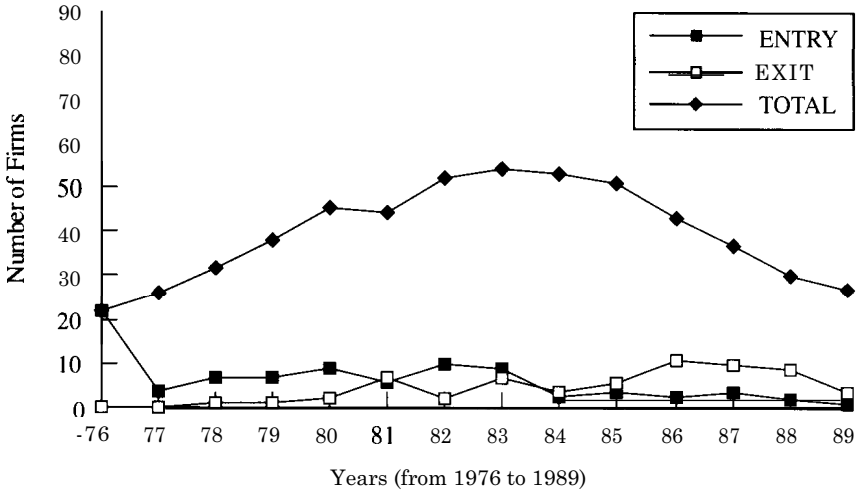
Clayton Christensen identifies five generations of Winchester drives in his data: 14-inch, 8-inch, 5¼-, 3½-inch, and 2½-inch drives. In the first two generations the motor was separate, while in the third it was incorporated in the drive spindle. Further important design changes were made in the 3½-inch drives, while the 2½-inch drives are essentially a miniature version of these and have not to date enjoyed important sales growth.[14] Christensen argues persuasively that established firms failed to master each successive generation of Winchester disk drive technology through being too wedded to existing customer demands and not sufficiently attentive to the emerging demands of smaller computer manufacturers.

Smaller drives were much slower and more expensive at first, but they did bring the hard drive to desktop machines. While established firms led the difficult but incremental improvement of components, new firms led with innovative architectures using existing components. The leaders in each generation of drives were different from those in the prior generation, with the exception of the current 2½-inch drives. Christensen claims that the 3½-inch drives incorporate all the features and functions that will be seen in 2½-inch and even smaller drives and thus constitute a dominant design. This is a pleasing conclusion because it perfectly supports the hypothesis with respect to the number of firms remaining in the industry, as shown by our independent analysis of his data in Figure 2-10.

Supercomputers

Supercomputers, the most powerful computational systems at any given time. today achieve speeds in the 100 MFLOPS (million floating operations per second) range. Three major technologies have been used to build these machines: sequential, vector, and parallel processing. Sequential computers, the architecture of which is often referred to as von Neumann, have only one central

FIGURE 2-10. **Number of Firms in the U.S. Winchester Disk Drive Industry**



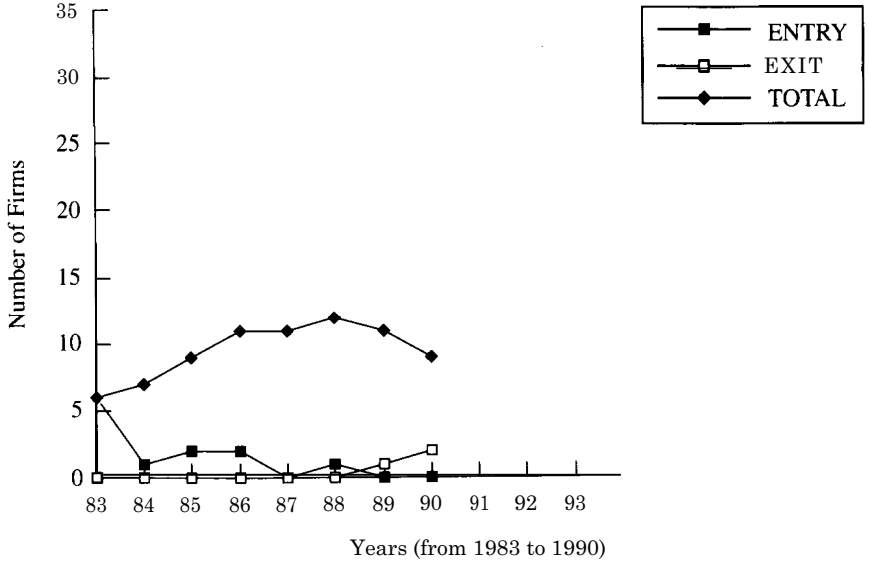
Source: Based on data from Christensen (see note 14).

processing unit (CPU); they do one thing at a time. Vector processors allow simultaneous computation for problems with vector-like or matrix-like structures. Parallel processing, or massively parallel processing, is a computer architecture in which hundreds or thousands of processors are put on many jobs simultaneously to finish tasks faster and with greater generality than do more traditional supercomputers. This, however, requires wholly new software.

Traditional supercomputer makers such as Cray, Fujitsu, Hitachi, IBM, and NEC produce mostly sequential machines. A second set of firms, minisupercomputer makers, now use sequential architecture with the incremental innovation of pipelining and vector processing. Massively parallel computer (MPC) makers are the latest entrants in the supercomputer industry, as shown in Figure 2-11. Firms such as Thinking Machines, Intel, Floating Point Systems, and Meiko started production about 1985, while the MPC pioneers (Ametek, Myrias, and Goodyear Aerospace) entered the industry only as far back as 1983.

If the MPC architecture becomes the dominant design, the exit of some traditional firms from the industry would seem likely in the future; firms such as IBM and Cray, which have greater resources, might turn toward this architecture.

FIGURE 2-1 1. **Number of Firms in the U.S. Massively Parallel Supercomputer Industry**



Other Research

Recently, Tushman and Anderson provided further evidence for the hypotheses considered here.[15] They gathered valuable data on the minicomputer, glass, and cement industries and performed tests of the model. Their work not only provides more data to support the concept of the dominant design and its importance in the life cycle of an industry, but also enhances our understanding of this process.

International Comparisons

The previous examples have illustrated how the entry and exit of firms from an industry parallels product innovation within that industry. Most of the examples provided here point to the conclusion that entering early is the most viable strategy. Entering at later stages, although theoretically possible, has proven to be a much riskier strategy and less likely to succeed. All of the evidence presented here, however, is drawn from the U.S. experience. The question arises: Does the pattern found in U.S. industry-that is, great falling off in the number of participating firms subsequent to the appearance of the dominant design-hold outside the United

States? A limited amount of research indicates that with respect to Japan it does not.

The Japanese experience with automobiles, calculators, integrated circuits, and Winchester disk drives shows an opposite pattern-of *more* firms joining these industries after the appearance of the dominant design. The successful late entry of large and highly integrated Japanese firms in some of our examples suggests that late entry, although difficult, is possible. The Japanese pattern would support the popular stereotype of American firms as innovators and creators of new industries and the Japanese as skillful at applying their considerable talents of automated production to product refinement and manufacture. However, the issue of late entry by Japanese firms could bear much further study. Early entry may still prove to be the strategy of choice in the future, even for Japanese firms, now that their research capabilities are world-class and the attention of American firms to excellence in manufacturing has been reawakened. In this sense, the strategies of firms in the two nations may be converging.

Limitations and Caveats

The idea of a dominant design does not seem to hold in industries producing nonassembled products such as rayon, glass, pulp and paper, metals, or industrial gases. Also, product categories such as integrated circuits and photographic film may be difficult to classify, as they share some characteristics of both assembled and nonassembled products.

Because nonassembled products contain fewer materials than assembled products do, there is more concentrated focus of technological effort and experimentation in the production process; periods of variation and experimentation result in what might be called an enabling technology. This enabling technology incorporates many of the elements needed in a continuous production process and allows the focus of technological effort to shift to process improvement rather than process innovation and design.

CAN WE RECOGNIZE A DOMINANT DESIGN WHEN WE SEE IT?

For managers, of course, the important question is whether a dominant design can be seen for what it is when it first appears or is discernible only in retrospect. If the emergence of a dominant

design is a signal that an important shift is about to take place with respect to the pace of innovation and the number of competing firms, the ability of managers and industry watchers to receive that signal-or even better, to predict the emergence of a dominant design-should be important with respect to their product designs, R&D, and process development. There are three schools of thought on this question.

First, some believe that dominant designs are the result of chance events. In this sense, their appearance cannot be predicted (though they may be recognized). This school would say that the QWERTY keyboard would be a curiosity piece today had August Dvorak and his ideas for keyboard design been on the scene earlier, during the period of experimentation with the first typewriters. Steel-bodied automobiles, the norm since Dodge's all-steel innovation of the 1920s, may have been dictated by the high cost of aluminum relative to steel at that time. Certainly aluminum, then as now, had tremendous performance advantages over heavier steel. Today, when the cost difference is not as great, the tradition of steel auto bodies is difficult to change.

The second school of thought has a deterministic viewpoint—i.e., something inherent in the technology determines the outcome with respect to the dominant design. The laws of nature, for example, fairly dictate that only a few synthetic materials (nylon, rayon, polyester) have chemical structures that support the spinning of long-fibered material. Thus the technological trajectory accommodates only a few possible candidates for the dominant design.

Third, others suggest that social and organizational factors work together to determine dominant design. Here again come into play arguments such as the early success of the QWERTY keyboard having created an investment in particular typing skills and barriers to other designs. The same may apply to the IBM PC; IBM's stature can be said to have led to an industry standard around which small competitors and software writers naturally gravitated.

Neither of these three schools of thought is entirely right or entirely wrong. Each, in fact, contributes to our understanding of the facts. But no matter how a dominant design is determined, it is doubtful that it can be recognized except in retrospect. Attempting to define or anticipate the appearance of a dominant design simply by mapping features and functions of the product alone is doomed to frustration. However, design simplicity and so-called technolog-

ical elegance are clearly characteristic of many dominant designs, the DC-3 aircraft being a notable example.

Our inability to immediately perceive a dominant design for what it is in no way diminishes its impact on the evolution of innovation and competition in an industry. For assembled products, as later chapters will show clearly, the appearance of a dominant design ushers in a period in which the rate of product innovation slows and the rate of process innovation increases.

DOMINANT DESIGNS AND THE COURSE OF FUTURE INNOVATION

A dominant design has other implications for business and R&D decision makers. As we have seen in the case of the typewriter, and as we will see in other case histories presented throughout this book, a dominant design has a powerful impact on the development of future technological innovations within the same industry. Discussion of the typewriter in the preceding chapter noted the extent to which this one innovation-a mechanical device for putting words on paper-altered life in the business workplace, with respect to both how work was done and who did it. An army of new workers emerged to handle the growing requirements for business letters, reports, and so forth. It is doubtful that the inventors of the typewriter anticipated this development. The standardization of typewriter design and the development of widespread QWERTY keyboarding skills created a market momentum that future innovations in typing, word processing, and personal computers were forced to recognize.

As new waves of innovation occurred—electric typewriters, stand-alone word processors, and personal computers-they did so within constraints of habits, skills, and user expectations set in place by the original innovation-the manual, QWERTY, Model 5-style typewriter. User acceptance of the dominant design of the original innovation created certain boundaries within which subsequent waves of innovation wisely developed. Thus IBM successfully sought to enhance performance of its electric typewriters through electromechanical applications only. Whether it would have enjoyed much market success if an improved keyboard configuration had been introduced at the same time-even if the keyboard was demonstrably more efficient-is doubtful. Likewise,

firms introducing new personal computers today must minimize their meddling with features in which current users have substantial investments: QWERTY keyboards, 3½-inch disk drives, operating systems, and so forth. Successful innovations in personal computers more often than not are transparent to users.

Later chapters will make it clear how the same phenomenon has occurred in other industries. For example, we will see in the next chapter how the displacement of gas lighting by incandescent lighting was advanced by Edison's running wires through the very same pipes that once brought illuminating gas into consumers' homes. Electric lamp sockets were cleverly adapted to former gas burners. Once the innovation of electric lighting established itself with users, subsequent technological innovations in lighting (fluorescent) and in electric appliances had to be made with the constraints imposed by the initial innovation (current, voltage, power grid, socket sizes, and so forth).

The lesson for technology managers and business strategists is straightforward: understand the constraints of systems, user learning, habits, and collateral assets already imposed by the existing dominant design if there is one. And as you develop innovative products, consider how these products will fit with current constraints.

ANTICIPATING MAJOR TECHNOLOGICAL CHANGES AND COMPETITIVE CHALLENGES

The preponderance of managers' external intelligence, thought, and action is usually directed toward competitors making products like their own that are similar in size, market approach, and competitive strategy. But as has been noted, the most threatening challenges are often those that come from outside the traditional definition of the industry and its products. Data presented in previous chapters suggest that architectural product innovations are often produced by new entrants-not only new firms, but also large enterprises entering a new area of business or spin-offs of established competitors formed to exploit a new idea. The formation of new firms is highly visible, and this suggests that their commitments to product introductions may be one of the most fruitful sources of early information about technological innovation in progress. One of the clear advantages of analyzing new firm activ-

ities is that there are far fewer of them than there are patents, technical papers, or variations of existing products-there is simply less noise to worry about in reaching a conclusion. This is the one sure approach that anyone could have used to identify the early implementation of xerography or of instant photography.

Twenty years ago computer memory was a wholly captive market, with a few large firms producing magnetic core memories for use in their own products. A survey conducted in 1970 by one of my students showed that all but five of 31 established computer companies were concentrating their resources on the manufacture of core memories. Twelve manufacturers outside the United States had a similar preponderance of magnetic core technology. However, eight newly formed firms exhibited a much different pattern. None were involved with magnetic core memories, while seven—including a struggling new company called Intel-were investing in development of the semiconductor memory chip. At least three of the new firms-Intel, Signetics, and Cogar-were spin-offs from larger firms and concentrated on more advanced technology than did their parents. Significantly, AT&T and IBM were among the larger firms also exploring the then-new memory chip idea. Analysis of the computer memory market as it unfolded provided a clear insight into the shape and direction that computer memory technology would take.

Monitoring new firm formation would have signaled clearly the vigor of the calculator and personal computer industries, the transistor and today's semiconductor memory chips, hard disk drives, and flash memories. The same would be true for the manual typewriter, the electric typewriter, and the word processor, as well as for minicomputers, fault-tolerant computers, supercomputers and massively parallel computers. This is not to say that new firms will be more successful financially than their established competitors. Being on the cutting edge of technology may be an extremely vulnerable position. As we have seen, many entrants in the race to start a new industry fail. The important point is that this is often the position taken by a newly formed firm.

A fascinating point to note in Table 2.2 is that older technologies may prefigure the most contemporary ideas. So-called flash memories, the new memory cards for portable computers, are present-day versions of the magnetic film memories first tried in the 1970s. Similarly, ceramic fibers used to insulate the space shuttle (among other applications) are in some respects an extension of the technology first tried in the production of ceramic filaments for gas lamps. Vacuum-tube design principles have been used to create a new system for spraying agricultural chemicals, reducing to one-tenth the chemicals earlier required and performing the job with inexpensive, small-scale equipment. And some of the current efforts to eliminate chlorofluorocarbons may benefit from knowledge of alternatives that were explored but abandoned earlier in favor of refrigerants now thought to be harmful to the earth's atmosphere. Thus looking at the evolution of technologies may help not only in anticipating possible directions of change, but also in thinking creatively about the design of new technologies.

During the past ten years scores of new firms have been formed to exploit findings in biotechnology, and new resource commitments are exceeding a billion dollars annually. With such widespread support and experimentation, one might anticipate almost the equivalent of a revolution in the fortunes of not only medicine and pharmaceutical firms, but also agriculture and the chemical and materials industries. Each innovation in biotechnology will be seen in terms of increasingly refined, enlarged, and more effective material forms long before it achieves widespread usage. The technical potentials in this area, though, will be much easier to assess than will be the business winners and losers. To anticipate com-

TABLE 2-2. U.S. Computer Memory Manufacturers' Product Offerings in 1970

Memory Technology	*Established Firms*	*New Firms*
Core memory	26	0
Plated wire	8	
Thin magnetic film	5	1
Semiconductor chips	6	7
Number of firms	31	8

Source: Compiled by the author from James M. Utterback and James W. Brown, *Business Horizons,* vol. 15 (October 1972).

mercial outcomes, one must look for indicators of combinations of economic and technical events as well as the political and social forces that may create either demand for or resistance to a new product or process-a far more complex task.

Notes

1. David Teece, "Profiting from Technological Innovation," *Research Policy* 15, no. 6 (1986). As Teece and others have noted, the dominant design model is better suited to mass markets, where consumer tastes are relatively homogeneous. Also, representations here about the dominant design extend only to assembled manufacturing industries. Among nonassembled products such as rayon or glass, innovation in the production process-not the emergence of a dominant product design-is a more powerful theme. (See Chapter 5 for further discussion.)
2. Teece, "Profiting from Technological Innovation."
3. Michael Cusumano, Y. Mylonadis, and R. Rosenbloom, "Strategic Maneuvering and Mass-Market Dynamics: The Triumph of VHS over Beta." *Business History Review,* vol. 66 (Spring 1992), 51-94.
4. Eric von Hippel, *The Sources of Innovation* (Oxford, England: Oxford University Press, 1988).
5. Joseph Schumpeter, *Business Cycles: A Theoretical, Historical, and Statistical Analysis of the Capitalist Process* (New York: McGraw Hill, 1939).
6. Burton Klein, *Dynamic Economics* (Cambridge, Mass.: Harvard University Press, 1977).
7. In fact, Edward B. Hess improved greatly on this aspect of keyboard mechanics by reversing the linkage between the key and the type bar, giving the machine a lighter touch and greater strike power for more even impressions.
8. George Nichols Engler, *The Typewriter Industry: The Impact of a Significant Technological Innovation,* Ph.D. diss. (Los Angeles: University of California at Los Angeles, 1970).
9. William J. Abernathy, *The Productivity Dilemma* (Baltimore, Md.: Johns Hopkins University Press, 1978), pp. 18-19.
10. R. Fabris, *Product Innovation in the Automobile Industry,* Ph.D. diss. (Ann Arbor: University of Michigan, 1966), pp. 85-93. These data are biased by the fact that Fabris counts only firms that produced automobiles for five years or more. Other, more comprehensive sources show massive entries and exits in the industry prior to 1923, though not all of the firms counted actually produced many cars.
11. I am grateful to John Rydz, a participant in the early RCA developments, for the information given here.
12. J.E. Tilton, *International Diffusion of Technology: The Case of Semiconductors* (Washington D.C.: The Brookings Institution, 1971).
13. William D. Bygrave and Jeffry A. Timmons, *Venture Capital at the Crossroads* (Boston: Harvard Business School Press, 1992), pp. 126-127. Fuller discussion of the skyrocketing growth and eventual implosion of the Winchester drive industry can be found in a case study by William A. Sahlman and Howard H. Stevenson, "Capital Market Myopia," 288–055 (Boston: Harvard Business School, 1987).
14. Clayton Christensen, *The Innovator's Challenge: Understanding the Influence of Market Environment on Processes of Technology Development in the Rigid Disk Drive Industry,* DBA diss. (Boston: Harvard Business School, 1992), pp. 42-73; also discussion with the author.
15. Michael Tushman and Philip Anderson, "Technological Discontinuities and Organizational Environment," *Administrative Science Quarterly 3* 1 (1986), 439–456.